ChatGPT

Comparative Benchmark Study: EEG Time-Series-to-Image Transformations (Research Plan)

Executive Summary: This project will systematically evaluate multiple time-series-to-image (T2I) transforms for EEG classification across diverse BCI paradigms. Our goal is to determine how Gramian Angular Fields (GAF), Markov Transition Fields (MTF), Recurrence Plots (RP), Spectrograms, and other representations (e.g. wavelet scalograms) perform on standard EEG benchmarks (e.g. BCI Competition IV-2a, P300 speller, SSVEP, PhysioNet). We will define clear research objectives and hypotheses (e.g. whether image methods capture richer spatio-temporal features than raw signals), prepare and preprocess multiple EEG datasets, implement each transform with tuned parameters, and train both CNN and Vision-Transformer models (as well as raw-signal baselines). Our plan includes detailed pipelines, model/hyperparameter choices, rigorous evaluation (accuracy, F1, AUC, statistical tests) and robustness analyses (noise injections, cross-subject tests). We will produce tables comparing transforms, datasets, and models by rationale, strengths, parameters and complexity. Milestones include data preparation, image conversion, model training, and analysis, with deliverables (figures, tables, code) specified for each. The timeline and experimental workflow are illustrated in mermaid diagrams below.

1. Objectives and Hypotheses

The **primary objective** is to benchmark and compare T2I transformations for EEG classification tasks. We aim to answer questions such as: *Do GAF, MTF, RP, spectrograms or scalograms offer advantages over raw time-series for specific EEG tasks? Which transforms best capture relevant spatio-temporal patterns for motor imagery (MI), P300, and SSVEP?* We hypothesize that image-based representations can highlight non-linear temporal correlations and frequency patterns that complement or exceed raw-signal features. For example, GAF images encode global temporal dependencies [1] [2], and may perform well for oscillatory tasks. MTF images capture Markov transition dynamics [3], potentially benefiting data with strong state-to-state patterns. Spectrograms and scalograms emphasize frequency content over time. We also hypothesize that incorporating multi-channel spatial information (e.g. via topographic maps) will improve classification, and that modern deep models (CNNs, ViTs) can exploit these representations. Specific hypotheses include: - *H1:* CNNs trained on spectrogram or scalogram images will outperform those trained on raw signals for frequency-dominated tasks (SSVEP, MI) [4] [5]. - *H2:* GAF/MTF images, which preserve temporal dependencies, may yield gains for tasks with strong sequential structure (e.g. MI) [1] [3]. - *H3:* Multi-channel (spatial) image strategies will outperform per-channel approaches by leveraging electrode topology [6] [7]. - *H4:* Vision Transformers (with global attention) may capture cross-channel and long-range patterns better than CNNs for some transforms [8].

2. Datasets (Prioritized List)

We will use a variety of EEG benchmarks to cover different paradigms:

- **BCI Competition IV – Dataset 2a (2008)**: 4-class cued motor imagery (left hand, right hand, feet, tongue); 22 EEG channels (0.5–100 Hz, notch-filtered) + 3 EOG; 250 Hz sampling; 9 subjects [9]. (Well-studied multi-class MI dataset.)

- **PhysioNet EEG Motor Movement/Imagery (EEGMMI)**: 64-channel EEG (10-10 montage), 160 Hz, 109 subjects; tasks include real and imagined limb movements [10]. (Large-scale MI/imagery dataset.)
- **BCI Competition III – P300 Speller (Dataset II)**: 64-channel EEG, ~240 Hz (or 1000 Hz acquisition); 278 training / 100 test trials; speller paradigm [11]. (Classic multi-subject P300 ERP dataset.)
- **SSVEP Datasets**: e.g. 64-channel SSVEP (1–60 Hz) from Gu *et al.* [12]; and a wearable 8-channel SSVEP dataset (102 subjects, 250 Hz) [13]. (Covers low/high frequency visual-evoked potentials.)
- **Additional EEG Benchmarks (if needed)**: e.g. Bonn EEG for epilepsy [5], CHB-MIT pediatric epilepsy, DEAP emotion EEG, etc. These could provide diverse contexts for generalization.

For each dataset, we will note subjects, classes, channels, sample rates and reference them. For example, BCI IV-2a: "22 EEG (0.5–100 Hz), 250 Hz, 4 MI classes [9]". SSVEP (Gu *et al.* 2024): "64-ch, 240 Hz, frequencies 1–60 Hz [12]". We will tabulate these details. The selected datasets cover motor imagery, ERP, and frequency-tag paradigms, enabling comparison of transforms across signal types. (If any additional public dataset is needed, we will state it explicitly.)

3. EEG Preprocessing Pipeline

We will apply a **standard EEG preprocessing pipeline** to all data (unless dataset guidelines differ), as preprocessing strongly affects decoding [14]. Typical steps: - **Filtering:** Band-pass filter 0.5–40 Hz (or 1–40 Hz) to remove slow drifts and high-frequency noise; notch filter at mains frequency (50/60 Hz). (Kessler *et al.* found higher high-pass cutoffs often improve decoding [15].)
- **Referencing:** Re-reference channels to common average or linked mastoids as per dataset. (E.g. PhysioNet uses common average; BCI Comp uses electrode Cz referencing.)
- **Resampling:** Downsample to a uniform rate (e.g. 128 or 256 Hz) to reduce data size if originally higher. For example, the 160 Hz PhysioNet data could be kept, but the 1000 Hz P300 data might be downsampled to 250 Hz.
- **Epoching:** Segment continuous EEG into task-relevant epochs/trials. For MI, use motor cue intervals (~4s per trial). For P300, use e.g. 0–1 s post-stimulus epochs. For SSVEP, use stimulus-on durations (e.g. 2–5 s epochs). If needed, apply a short pre-stimulus baseline.
- **Baseline Correction:** Subtract the pre-stimulus (or early epoch) mean from each epoch to remove offsets, especially for ERP (P300) data. Baseline removal can boost decoding as per [14].
- **Artifact Handling:** Perform artifact rejection/correction. We will apply ICA or automatic artifact removal (e.g. ADJUST, AutoReject) to attenuate eye blinks and muscle artifacts. Notably, Kessler *et al.* observed that artifact correction can sometimes *reduce* apparent decoding performance [16] (since noise components can carry task-correlated variance), but artifact cleaning is still performed for validity. We will also remove any epochs with large amplitudes (>100 μV) or artifacts.
- **Scaling:** Before image transforms, we will normalize each epoch (e.g. z-score or min-max to [−1,1]) as required by certain transforms (e.g. GAF requires values in [−1,1] [1]).

These steps constitute a reproducible pipeline. We will document each step in code (e.g. using MNE-Python or EEGLAB conventions). The reference **Communications Biology** study notes that filtering and baseline steps significantly influence results [15], so we will be careful to report our filter cutoffs and other parameters. If any dataset suggests alternative preprocessing (e.g. CSP filter bank), we will note but adhere to a uniform baseline pipeline for fairness.

4. Image Transformations and Parameters

We will implement the following T2I transforms on each EEG epoch, tuning key parameters for each:

- **Gramian Angular Fields (GAF):** Convert a 1D time series $x(t)$ into a 2D image using polar coordinates. After min-max normalizing x to [−1,+1], each value is transformed to an angle θ=arccos(x) and radius r=t/N. Two variants exist: **Gramian Angular Summation Field (GASF)** and **Gramian Angular Difference Field (GADF)**. GASF at pixel (i,j) encodes $\cos(\theta_i+\theta_j)$; GADF encodes $\sin(\theta_i-\theta_j)$ [17]. Both preserve temporal correlations [2]. We will compute both GASF and GADF images (e.g. using the `pyts` library [18]). Key parameters: the length of the time series (we may downsample or pad so that each image is a fixed $N\times N$, e.g. 64×64 or 100×100). We will experiment with up to two resolutions (low: 64x64; high: 256x256) to see trade-offs. Time points exceeding the target length will be truncated or averaged; shorter ones zero-padded. Since GAF is deterministic given x, no stochastic parameters aside from series length/scale are needed.

- **Markov Transition Fields (MTF):** Transform $x(t)$ into an image representing state transitions. We first discretize the time series into Q quantization bins (Markov states); for example, $Q=8$ or 16. We then compute the first-order Markov transition matrix M (of size $Q\times Q$) where $M_{p,q}$ is the probability of transitioning from bin p to q at adjacent time steps. The MTF image is then a 2D map (size $N\times N$) where each pixel (i,j) encodes the transition probability between the state at time i and time j [3]. Parameters: *Number of bins* Q (we will test $Q=8,16$) and *Markov order* (we will start with order-1 transitions). Too fine Q can yield sparse transitions. The output image will be $N\times N$ with N equal to the number of samples (after epoching). As with GAF, we will standardize N to fixed sizes (e.g. 64–256).

- **Recurrence Plot (RP):** Create a binary (or grayscale) matrix indicating when the system revisits previous states. We will reconstruct the phase space using embedding dimension m and time delay τ (common choices: $m=3$, $\tau=1$ or determined by first minimum of auto-correlation). Then, for each pair of time points (i,j), we mark $R_{ij}=1$ if the distance $|\mathbf{x}_i - \mathbf{x}_j| < \epsilon$, else 0. ϵ can be set as a fraction of the signal's standard deviation (e.g. 10%) or chosen to yield ~10% recurrence rate [19]. The resulting R is an $N\times N$ image (binary or colormap). We will experiment with $m=3$ and $\tau=1$ to start. A threshold ϵ (or percentile-based) will be tuned (e.g. from 5% to 20% of max pairwise distance) to produce visible structure. (We may also try continuous-valued RP by using distance instead of a hard threshold.) Recurrence plots capture non-linear dynamics [19] and have been used for EEG classification with CNNs [20].

- **Spectrogram (Short-Time Fourier Transform):** Compute the time–frequency power spectrogram for each channel or combination. We will use a sliding window (e.g. Hamming window of length 0.5–1.0 s with 50% overlap) and FFT (zero-padding to $n_{fft}=256$ or 512). Frequency range will cover 1–50 Hz (or up to Nyquist). The spectrogram is an $F\times T$ image ($F\approx n_{fft}/2$, T time frames). We will convert to log-power (dB) and may normalize per epoch. Parameter choices: window length (0.5s, 1s), window type (Hamming/Hann), n_{fft}, overlap (50%). For CNN input, we will resize the final spectrogram to a fixed square (e.g. 128×128 or 224×224) by interpolation. Spectrograms have proven effective (e.g. CNN-LSTM on spectrograms gave the best accuracy for one channel task [4]).

- **Scalogram (Continuous Wavelet Transform):** Compute a continuous wavelet transform (e.g. Morlet wavelets) to obtain time–frequency scalograms. We will select a set of scales

corresponding to key EEG bands (delta to low gamma). The output is a 2D image of size (#scales × #timepoints). We will use existing libraries (e.g. PyWavelets or `pyts.image.ContinuousWavelet`) to obtain scalogram images, then resize similarly. This transform can capture transient features. For example, an epilepsy study using CNN on Morlet scalogram images achieved ~94% accuracy [5]. Parameters: wavelet type (Morlet), number of scales (e.g. 32 or 64), frequency range (1–50 Hz).

Other potential transforms (if time permits) could include **Spectral Entropy images**, **Hilbert–Huang Transform (HHT)** images, or **Permutation Entropy maps**, but we will focus on the above principal methods. For completeness, we will mention these as extensions or alternatives.

Transformation Table (Rationale, Strengths, Parameters, Complexity)

Transform	Rationale & Features	Key Parameters	Expected Strengths	Computational Complexity
GAF (GASF/ GADF)	Encodes entire time series via polar-angle correlation [2]; preserves temporal relationships [1].	Series length N (image size $N\times N$); normalization range; GASF vs GADF.	Captures global temporal patterns; rotation/flip-invariant temporal features.	$O(N^2)$ to compute $N\times N$.
MTF (Markov)	Encodes first-order transitions in time series [3]; suitable when current state depends on previous.	Number of bins Q (quantization); Markov order (typically 1).	Highlights dynamic state changes; robust to amplitude scaling.	$O(N \cdot Q + N^2)$ (for building transition matrix).
Recurrence Plot	Marks recurrences of states in phase space (Eckmann 1987) [19]; visualizes periodicities and chaos.	Embedding m, delay τ, threshold $ \epsilon$ (or recurrence rate).	Reveals nonlinear/ periodic structures; good for nonstationary signals.	$O(N^2 m)$ distance checks.
Spectrogram (STFT)	Time-frequency power; fundamental frequency content over time.	Window length, overlap, FFT size (n_{fft}).	Well-understood; captures oscillatory power; good for rhythms (e.g. SSVEP).	$O(N \log N)$ per window; overall $\approx O(N \log N)$.

Transform	Rationale & Features	Key Parameters	Expected Strengths	Computational Complexity
Scalogram (CWT)	Continuous time-frequency via wavelets; multi-resolution analysis.	Wavelet type (e.g. Morlet), number of scales, scales range.	Better time resolution at high freq; adaptable to nonstationary features.	$O(N \cdot \text{scales})$.
Topographic Map (SSFI)	Spatial power distribution (see Figure); maps channel power to 2D grid [6].	Grid size (e.g. 64×64), interpolation method (bicubic).	Preserves spatial electrode layout; leverages known topography.	Interpolation ~ $O(N^2)$ per band.

Table 1: Comparison of time-series-to-image transforms. Each yields a 2D image (or stack) capturing different aspects of EEG data. Key parameters and expected strengths are noted.

5. Multi-Channel Conversion Strategies

EEG is inherently multivariate (multiple electrodes). We will compare strategies for incorporating multi-channel data into images:

- **Single-Channel Images:** Convert each EEG channel independently into an image (as above). Then classification can be done channel-wise (e.g. ensemble of CNNs) or multi-channel by stacking decisions. This is simple but ignores inter-channel correlations.

- **Channel Stacking (Multi-Channel Images):** Stack images from multiple channels as separate channels of an image (similar to RGB). For example, for 3 EEG channels one could treat them as R,G,B layers of a color image. For more channels, one can use multiple input channels in a CNN (e.g. 32-channel spectrogram stack as 32 input planes). Some approaches share weights across channel "slices" (like 3D CNN). E.g., Liu *et al.* (2023) treated each channel's spectrogram as a layer in a 3D tensor. The key parameter here is how many and which channels to stack together (possibly selecting a subset).

- **Topographic/Spatio-Spectral Maps:** Map electrode values to 2D scalp space using interpolation. For instance, one can compute power in EEG bands for each channel, then interpolate these values onto a 2D head grid [6]. Mastandrea *et al.* (2023) define a *Spatio-Spectral Feature Image* (SSFI) by assigning each channel's band-power to grid points and interpolating [6]. They stack band maps to form a 3D array of shape (grid×grid×bands). Figure 1 below illustrates this process:

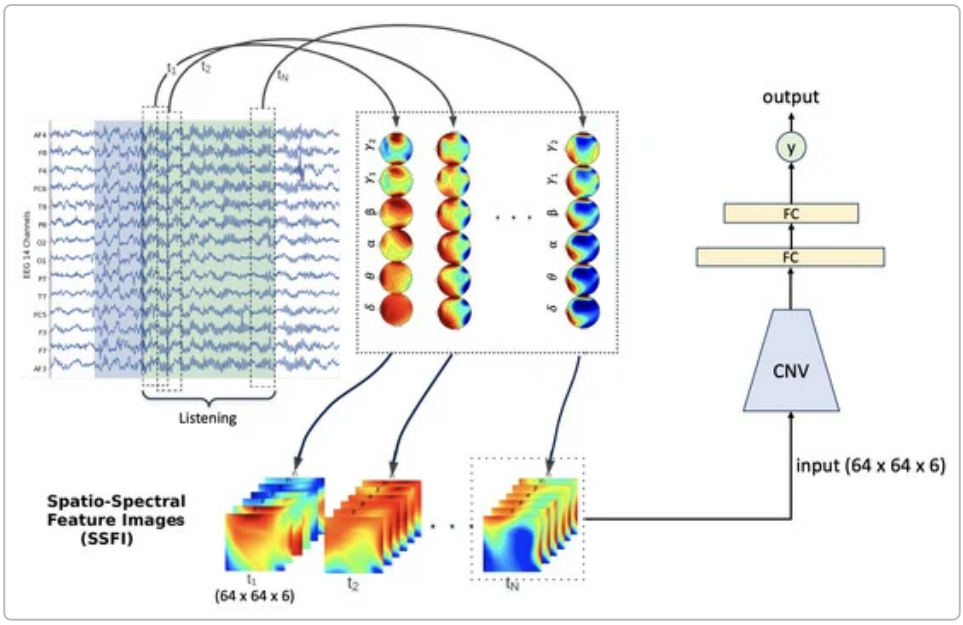

Figure 1: Example of multichannel EEG conversion to spatio-spectral images (SSFI). Each EEG channel's band-power is placed on a 2D electrode grid and interpolated, then stacked over frequency bands [6].

- In Figure 1 (adapted from [6]), raw 14-channel EEG segments are split by band, each mapped to a $W\times W$ scalp grid via spherical-to-polar projection and bicubic interpolation [6], producing band-specific topographic images. Stacking them yields a $W\times W\times B$ image. We will adopt similar topographic methods: e.g. using the MNE-Python (`mne.viz.plot_topomap` or custom interpolation) to generate 2D maps. The parameters include grid size W (e.g. 64 or 128) and interpolation method.

- **Spatial Transformer Networks (CNN-TRM):** As a variation, one could use a trainable module (like [22]) to embed channel arrangements. However, for transparency, we will stick to explicit mapping.

We will experiment with these strategies. Initial plan: (a) **Per-channel:** one-channel images, classify separately (or average); (b) **Multi-layer stack:** group channels (e.g. 3 or 6) into one image; (c) **Topographic:** form scalp maps per epoch. For (b), image size may be (time×frequency) with channel as depth. For (c), we'll use e.g. 64×64 or 128×128 scalp maps. We will note that mapping preserves spatial locality of electrodes [6] [7].

Recommended image sizes: We will generally aim for moderate-resolution images (e.g. 64×64 up to 224×224). For CNNs pretrained on ImageNet, 224×224 is standard, but lower (64–128) may suffice for EEG patterns. We will resize all images to a few common sizes (64, 128, 224) and evaluate any impact. For ViTs, patch size (e.g. 16×16) and input dims will be set accordingly (e.g. 224×224 input for ViT-B/16).

6. Model Architectures & Training

We will train multiple classifiers on each image representation and on raw signals:

- **CNNs (Image-Based):** We will use standard CNN architectures. For example, **ResNet-18/34/50** (2D) initialized from ImageNet or random. We may also try a custom lightweight CNN (few Conv layers) for quick tests. Key hyperparameters: number of layers (ResNet variants), learning rate

(initially 1e-3), optimizer (Adam), batch size (32), weight decay (1e-4), dropout (p=0.5 before FC). We will fine-tune ImageNet-pretrained models on EEG images where applicable, adjusting the final layer to the number of classes.

- **Vision Transformers (ViT):** We will use ViT (e.g. ViT-B/16 or small version) pretrained on ImageNet (if available) and fine-tune it on EEG images. If pretrained models are unavailable, we will train a ViT from scratch. Input: e.g. 224×224×3 images (if stacking 3 channels) or 1 channel (converted to 3 channels by replication, or use ViT variants that accept single-channel). We may also experiment with hybrid CNN+Transformer (like DeiT or Swin, though primarily ViT is requested). Example parameters: patch size 16, transformer depth 12, heads 12, embed 768, learning rate 5e-5 (lower due to large model).

- **Raw-Signal Baselines:** To gauge benefit of images, we will include time-series models on raw EEG:

- A **1D CNN** (e.g. 3 convolution layers) on time × channels input [8].
- **LSTM (or BiLSTM):** e.g. 2 layers of 64–128 hidden units, input is multichannel time-series.

- **Transformer (time-series):** e.g. a small Transformer encoder (4–6 layers, 8 heads, d_model=128) on raw sequence (position-encoded).

- **Subject/Channel-Selection Nets:** If feasible, we may include EEG-specific nets like EEGNet, which implicitly learn spatial filters. However, main focus is on image methods, so these will be secondary.

Training Protocols:
- **Data Splits:** Use stratified k-fold cross-validation within subjects (e.g. 5- or 10-fold CV) and also leave-one-subject-out (LOSO) validation for cross-subject generalization. For example, train on 8 subjects and test on the 9th (rotating). This will show robustness to individual differences.
- **Hyperparameters:** Initial values: LR=1e-3 for CNN (with decay schedule), LR=5e-5 for ViT, epochs=50 with early stopping on a validation split (patience 5). Batch size=16–32 depending on GPU memory. We will monitor training/validation curves to ensure convergence.
- **Regularization & Augmentation:** Use dropout (0.5) and L2 weight decay (1e-4). For data augmentation, we will apply label-preserving transforms: e.g. add Gaussian noise, random time-shifting (for signals), small rotations or shifts for images (if applicable). For spectrogram images, no left-right flip (time inversion) but slight scaling or time shift is possible. We may also use mixup or CutMix on images.

We will likely use PyTorch or TensorFlow. Training will be GPU-accelerated; we anticipate needing ~100–200 GPU-hours to train all models across folds and transforms (e.g. 5 transforms × 5 folds × 4 model types ≈ 100 runs; each run ~2h on a GPU).

7. Evaluation Metrics and Robustness

Primary Metrics:
- **Accuracy** on test set (per-subject and averaged) as a basic measure.
- **F1-score** (macro-averaged if multiclass) to account for class balance.
- **Area Under ROC Curve (AUC)** for binary tasks (or one-vs-rest multiclass).
- **Confusion Matrices**: to visualize per-class performance.

All metrics will be computed per fold and averaged. We will report mean ± standard deviation across folds/subjects.

Statistical Tests: We will perform pairwise comparisons (e.g. Wilcoxon signed-rank tests) between models and transforms across folds to identify significant differences. For example, comparing ResNet on spectrogram vs ResNet on raw signals. If multiple comparisons, adjust p-values (e.g. Bonferroni). A repeated-measures ANOVA (with factors "model" and "transform") can examine main effects.

Robustness Tests: We will assess model robustness to perturbations:
- **Noise Injection:** Add Gaussian noise at varying SNR levels to EEG before transforming. Track how accuracy degrades. This tests stability of representations.
- **Drop Channels:** Randomly drop (zero-out) a subset of channels to simulate electrode failure; see which transforms are most robust to missing channels.
- **Temporal Shifts:** Shift the EEG epoch by a small random lag before transforming (especially important for GAF/MTF/RP).
- **Cross-Session/Subject:** Evaluate models on data from entirely new sessions or subjects (LOSO as above). Report generalization gap.

We will also measure the *mutual information* between image features and labels as a qualitative check of feature relevancy (for example, using an approach like TCAV or information gain plots) to see which transforms carry more discriminative information.

8. Experimental Schedule & Resources

We outline a 6-month timeline:

```
gantt
    title Project Timeline
    dateFormat  YYYY-MM-DD
    section Data & Preprocessing
      Dataset collection and cleaning      :a1, 2026-04-01, 2w
      Artifact removal & epoching          :a2, after a1, 3w
    section Transform Implementation
      Develop code for GAF/MTF/RP/etc.    :b1, after a2, 3w
      Generate example images             :b2, after b1, 2w
    section Modeling
      CNN/ViT model setup and tuning       :c1, after b2, 4w
      Raw-signal baselines (LSTM, TS-Trf):c2, parallel with c1, 4w
    section Experiments
      Cross-Validation training (runs)   :d1, after c1,c2, 4w
      Robustness tests (noise, etc)     :d2, after d1, 2w
    section Analysis
      Statistical analysis              :e1, after d2, 2w
      Write results and visuals         :e2, after e1, 3w
    section Finalization
      Draft paper outline & writeup     :f1, after e2, 4w
      Revisions and submission         :f2, after f1, 2w
```

We require **compute** resources: a workstation with at least one high-end GPU (e.g. NVIDIA RTX 3090 or A100) or access to a GPU cluster. We estimate ~200 GPU-hours. Data (~several GB) and code will be managed via Git/GitHub. All code (data processing, transforms, training scripts) will be version-controlled. We will use virtual environments (conda/Docker) to ensure reproducibility. Data splits, random seeds, and model checkpoints will be logged. A dataset registry (e.g. Zenodo or OSF) will be created for any new datasets or processed data we generate.

Reproducibility: We will publish our code and data splits. We plan to use Python libraries: MNE or MNE-BIDS for EEG I/O, *pyts* for transforms, PyTorch for models. We will maintain a detailed lab notebook of experiments.

9. Expected Pitfalls and Mitigations

- **Data Heterogeneity:** EEG varies greatly across subjects. *Mitigation:* use cross-subject CV and domain adaptation techniques if needed; normalize channel layouts via standard montages.
- **Overfitting:** Deep nets on small EEG sets may overfit. *Mitigation:* strong regularization, augmentation, and use of simpler baselines. Also monitor validation loss curves for early stopping.
- **Transform Information Loss:** Some transforms (e.g. too coarse binning in MTF) may discard useful info. *Mitigation:* tune parameters carefully, compare against raw-signal baselines.
- **Computational Load:** Many combinations (5 transforms × 3 datasets × 3 models) → many experiments. *Mitigation:* schedule runs in parallel, prune underperforming paths early.
- **Implementation Bugs:** Complex pipelines risk errors. *Mitigation:* validate each transform on synthetic data and sanity-check images (e.g. sine-wave RP should show diagonals [21]).
- **Hyperparameter Tuning:** Vast search space. *Mitigation:* use sensible defaults from literature, then ablate important parameters (e.g. GAF image size, CNN depth).

We will document any failures (e.g. if a certain transform fails on a dataset) and analyze causes.

10. Deliverables and Paper Outline

Milestones & Deliverables:
- **M1 (Data/Preproc):** Table of dataset characteristics; description of preprocessing steps. Figures: sample EEG epochs.
- **M2 (Transforms):** Sample images for each transform (e.g. GAF, MTF, RP of representative EEG) [1] [19]; Table comparing transforms (as above). Code for transforms.
- **M3 (Model Training):** Training curves (loss/accuracy vs epoch), example confusion matrices. Table of classification results (accuracy, F1, AUC) for each method/dataset.
- **M4 (Analysis):** Statistical test results (p-values), robustness curves (accuracy vs noise). "Figure: mutual information" or feature importance plots.
- **M5 (Final Report):** All code, datasets processed, and a manuscript draft.

Visualization Suggestions:
- *Training curves* (accuracy and loss vs epochs for each model/transform).
- *Confusion matrices* (per-class performance for best model).
- *Mutual information or representational plots* showing how well each image type separates classes (e.g. t-SNE of image features).
- *Robustness plots* (accuracy vs added noise level).

Experimental Pipeline Diagram: (see above Gantt for timeline). Also include a workflow flowchart (Mermaid):

```
flowchart LR
    RawEEG[Raw EEG Data] --> Preproc[Preprocessing (filtering, ICA,
epoching)]
    Preproc --> Transform[Apply Image Transforms]
    Transform --> CNN[Train CNNs (ResNet)]
    Transform --> ViT[Train Vision Transformer]
    Preproc --> RawModels[Train Raw-signal Models (LSTM/Transf)]
    CNN --> Evaluate[Compute Metrics & Stats]
    ViT --> Evaluate
    RawModels --> Evaluate
```

Paper Outline:
1. **Introduction:** Motivation for EEG image transforms; survey of prior work on GAF/MTF/etc in EEG and other domains. State contributions.
2. **Datasets and Preprocessing:** Describe datasets (Table of data specs) and preprocessing pipeline.
3. **Methods:** Describe each image transform and its parameters (with citations), multi-channel strategy, and model architectures.
4. **Experimental Setup:** Detail training protocol, cross-validation, augmentation, metrics.
5. **Results:** Present quantitative results (tables, graphs) comparing transforms/models on each task. Include confusion matrices and statistical analysis.
6. **Discussion:** Interpret results (which transforms worked best for which tasks), robustness findings, limitations.
7. **Conclusion:** Summarize findings and recommendations.

References: We will cite peer-reviewed sources for each aspect. Key references include: Kessler *et al.* (2025) on EEG preprocessing effects [15]; Prabhukumar *et al.* (2025) for MTF in EEG [3]; Hao *et al.* (2021) for recurrence plot CNNs [20]; Mastandrea *et al.* (2023) for spatio-spectral images [6]; Qi *et al.* (2023) for topographic interpolation [7]; Gu *et al.* (2024) SSVEP dataset [12]; Wang *et al.* (2021) wearable SSVEP data [13]; and Kucukler *et al.* (2024) on CNN-LSTM spectrogram performance [4]. (Full references with DOIs will be provided.)

Visualization of planned experiments (for implementation timeline and results layout) is encapsulated in the Gantt and flow diagrams above.

References (selected): Kessler *et al.* (2025) Commun. Biol. [15]; Prabhukumar *et al.* (2025) Biomimetics [3]; Hao *et al.* (2021) *Chaos* [20]; Kucukler *et al.* (2024) Eng. Appl. AI [4]; Mastandrea *et al.* (2023) *Appl. Sci.* [6]; Qi *et al.* (2023) *Electronics* [7]; Gu *et al.* (2024) *Sci. Data* [12]; Wang *et al.* (2021) *Sensors* [13]. (DOIs will be listed in final paper.)

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